Strategic Behavior

Final Exam: Solutions and Grading Guide • Spring 2024

Dr. K. Osei

General grading policy. Grade each step as listed. Carry forward a student's numerical slip: if later work is correct given the wrong number, award those later points. For verbal interpretations, award credit only when the direction of the effect matches the student's computation. Monetary amounts in Question 1 are in thousands of dollars and in Question 5 in millions of dollars.

Question 1: The Tractor Auction (20 points)

(a) Fix a collector with value v and let B be the highest competing bid. Bidding v yields $v-B$ if $v>B$ and 0 if $v<B$ (and 0 in expectation at a tie $B=v$), which is $\max\{v-B,0\}$.

- Bidding $b>v$ changes the outcome only when $v\le B<b$: the bidder now wins and pays $B\ge v$, earning $v-B\le 0$ instead of 0.
- Bidding $b<v$ changes the outcome only when $b<B<v$ (or at ties): the bidder now loses and earns 0 instead of $v-B>0$.

So for every B truthful bidding does at least as well as any other bid, and strictly better for some B: it is weakly dominant. The price paid never depends on one's own bid, only whether one wins.

> **Grading.** 1 pt: setup with B. 1.5 pts: overbidding case. 1.5 pts: underbidding case. An argument using only "you pay the second price so you can't gain" without the two cases: 2/4.

(b) Suppose the other three collectors bid $\frac{3}{4}V_j$ with $V_j\sim U[0,60]$. Their bids lie in $[0,45]$. A bid $b\ge 45$ wins for sure, so any bid above 45 pays more than 45 without raising the chance of winning; we may restrict to $b\in[0,45]$. A bid b beats one rival with probability $\Pr(\frac{3}{4}V_j<b)=\frac{4b}{3\cdot 60}=\frac{b}{45}$, so

$$\Pi(b)=(v-b)\Big(\frac{b}{45}\Big)^3,\qquad \Pi'(b)=\frac{b^2}{45^3}\big[3(v-b)-b\big]=\frac{b^2}{45^3}\,(3v-4b).$$

$\Pi'>0$ for $b<\frac{3}{4}v$ and $\Pi'<0$ for $b>\frac{3}{4}v$, so $b=\frac{3}{4}v\in[0,45]$ is the global maximizer. Hence $b(v)=\frac{3}{4}v$ is a best response to itself: a symmetric BNE.

> **Grading.** 1 pt: bids above 45 dominated. 2 pts: winning probability $(b/45)^3$. 2 pts: FOC gives $b=\frac{3}{4}v$. 1 pt: global optimality (sign of Π'). Guessing the formula $\frac{n-1}{n}v$ without verification: 1/6.

(c) With $n=4$, $E[\text{highest}]=\frac{4}{5}\cdot 60=48$ and $E[\text{second highest}]=\frac{3}{5}\cdot 60=36$.

- Second-price: revenue $=E[\text{second highest value}]=$ 36 .
- First-price: revenue $=E\big[\frac{3}{4}\cdot\text{highest value}\big]=\frac{3}{4}\cdot 48=$ 36 .

The two auctions raise the same expected revenue, as the revenue equivalence theorem predicts.

> **Grading.** 2 pts each auction. 1 pt: mentions revenue equivalence (or equality). Using $E[\text{highest}]$ for the second-price auction: 0 for that part.

(d) Rescale to $U[0,1]$ values (multiply by 60 at the end); the reserve becomes $\frac{1}{2}$. Revenue comes from two events:

- Exactly one value is at least $\frac{1}{2}$: probability $4 \cdot \frac{1}{2} \cdot \left(\frac{1}{2}\right)^3 = \frac{1}{4}$; the price is $\frac{1}{2}$. Contribution $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$.
- The second-highest value $X_{(2)}$ is at least $\frac{1}{2}$; the price is $X_{(2)}$. Contribution

$$\int_{1/2}^{1} x \cdot 12x^2(1-x)\,dx = 12\Big[\frac{x^4}{4} - \frac{x^5}{5}\Big]_{1/2}^{1} = 12\Big(\frac{1}{20} - \frac{3}{320}\Big) = \frac{39}{80}.$$

Expected revenue $= 60\big(\frac{1}{8} + \frac{39}{80}\big) = 60 \cdot \frac{49}{80} = 7.5 + 29.25 =$ 36.75, higher than 36, even though the tractor now goes unsold with probability $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$. The reserve extracts more from the high bidder when competition is thin.

> **Grading.** 2 pts: event "exactly one above r" and its contribution 7.5. 2 pts: the integral and 29.25. 1 pt: total 36.75 with comparison to 36.

Question 2: Deterring Entry into the Ready-Mix Market (18 points)

(a) If E enters, it maximizes $(90 - q_I - q_E)\,q_E$, so $q_E = \dfrac{90 - q_I}{2}$ (for $q_I \le 90$). Then the price is $\frac{90-q_I}{2}$ and E's profit from entering is

$$\pi_E(q_I) = \Big(\frac{90 - q_I}{2}\Big)^2 - F\,.$$

> **Grading.** 2 pts: q_E. 2 pts: profit (including $-F$).

(b) Firm I anticipates $q_E = \frac{90-q_I}{2}$ and maximizes $q_I\big(90 - q_I - \frac{90-q_I}{2}\big) = \frac{1}{2}q_I(90 - q_I)$, giving $q_I = 45$. Then $q_E = 22.5$, $P = 22.5$, $\pi_I = 45 \cdot 22.5 = 1012.5$ and $\pi_E = 22.5^2 = 506.25$.

> **Grading.** 2 pts: $q_I = 45$ via backward induction. 1 pt: q_E. 1 pt: price. 1 pt: profits. Solving the simultaneous Cournot game instead ($q = 30$ each): 1/5.

(c) $F = 225$. Entry is deterred iff $\left(\frac{90-q_I}{2}\right)^2 \le 225 \iff q_I \ge 60$, so the limit output is $q_I = 60$.

- Deter: for $q_I \ge 60$ firm I is a monopolist with profit $q_I(90 - q_I)$, which is decreasing for $q_I > 45$; the best choice is $q_I = 60$ with profit $60 \cdot 30 = 1800$.

- Accommodate: for $q_I < 60$ entry occurs and the best choice is the Stackelberg output $45 < 60$, with profit 1012.5.

Since $1800 > 1012.5$, firm I **deters**: SPE outcome $q_I = 60$, no entry, $P = 30$, $\pi_I = 1800$, $\pi_E = 0$.

> **Grading.** 2 pts: limit output 60. 2 pts: comparison 1800 vs. 1012.5. 1 pt: outcome. Students who say "deter by producing the monopoly output 45": 0 for the limit output.

(d) $F = 25$. Deterrence requires $\frac{90-q_I}{2} \le 5 \iff q_I \ge 80$, which yields $80 \cdot 10 = 800 < 1012.5$. Firm I **accommodates**: $q_I = 45$, E enters with $q_E = 22.5$, $P = 22.5$, $\pi_I = 1012.5$, and $\pi_E = 506.25 - 25 = 481.25$.

> **Grading.** 1 pt: limit output 80. 1 pt: deterrence profit 800. 2 pts: decision and outcome, including E's net profit.

Question 3: Can the Cement Cartel Hold? (20 points)

(a) $\pi_i = (120 - q_i - q_j)\,q_i$ gives $BR_i = \frac{120-q_j}{2}$. The symmetric solution is $q_i = 40$, $P = 50$, profit $40 \cdot 40 =$ 1600 each.

> **Grading.** 2 pts: best response. 1 pt: quantities. 1 pt: profits.

(b) The monopoly maximizes $(120 - Q)Q$: $Q = 60$, $P = 70$. Each firm produces 30 and earns $60 \cdot 30 =$ 1800.

> **Grading.** 2 pts: monopoly output. 1 pt: per-firm profit.

(c) If the rival produces 30, the best deviation is $q = \frac{120-30}{2} = 45$, giving $P = 55$ and profit $45 \cdot 45 = 2025$. Grim trigger is an SPE iff

$$\frac{1800}{1-\delta} \;\ge\; 2025 + \frac{1600\,\delta}{1-\delta} \iff 1800 \ge 2025 - 425\delta \iff \delta \ge \tfrac{9}{17} \approx 0.53\,.$$

The punishment phase (Cournot forever) is a repetition of the stage NE, so no deviation from it is profitable.

> **Grading.** 2 pts: deviation quantity and profit. 3 pts: incentive constraint. 1 pt: $\frac{9}{17}$. 1 pt: punishment phase is credible.

(d) With three firms:

- Cournot: $BR_i = \frac{120-Q_{-i}}{2}$; symmetric $q = 30$, $P = 40$, profit 900.

- Collusion: $Q = 60$ split three ways, $q = 20$, profit $60 \cdot 20 = 1200$.
- Deviation against $20 + 20$: $q = \frac{120-40}{2} = 40$, $P = 50$, profit 1600.

$$\frac{1200}{1-\delta} \geq 1600 + \frac{900\,\delta}{1-\delta} \iff 1200 \geq 1600 - 700\delta \iff \delta \geq \tfrac{4}{7} \approx 0.57 .$$

Since $\frac{4}{7} > \frac{9}{17}$, entry makes collusion **harder** to sustain: each firm's share of the cartel profit shrinks while the gain from cheating stays large.

> **Grading.** 1 pt each: Cournot, collusion, deviation values. 2 pts: threshold $\frac{4}{7}$. 1 pt: comparison and explanation.

Question 4: Funding the Community Garden (22 points)

(a) Given G_{-i}, neighbor i maximizes $-g_i + a_i \ln(G_{-i} + g_i)$, a strictly concave function of g_i. The first-order condition $a_i/(G_{-i} + g_i) = 1$ gives $G = a_i$ when interior; with the constraint $g_i \geq 0$,

$$g_i(G_{-i}) = \max\{a_i - G_{-i}, 0\} .$$

Neighbor i tops up the total to a_i if others give less, and gives nothing otherwise.

> **Grading.** 2 pts: FOC. 2 pts: corner at 0. 1 pt: concavity or second-order check.

(b) In any equilibrium $G > 0$. A neighbor with $g_i > 0$ must have $G = a_i$, and a neighbor with $g_j = 0$ must have $G = G_{-j} \geq a_j$. Two neighbors cannot both contribute, since that would require $G = a_i = a_j$ while the a's are distinct. So exactly one neighbor i contributes, $G = a_i$, and $a_i \geq a_j$ for all j; hence $i = 1$. The unique Nash equilibrium is

$$(g_1, g_2, g_3) = (30, 0, 0), \quad G = 30 .$$

Check: neighbors 2 and 3 face $G_{-j} = 30 \geq a_j$, so they contribute nothing.

> **Grading.** 2 pts: conditions for contributors/non-contributors. 2 pts: at most one contributor. 2 pts: identifies neighbor 1 and $G = 30$ with verification.

(c) The sum of payoffs is $3w - G + (a_1 + a_2 + a_3)\ln G$. Maximizing gives $\frac{a_1+a_2+a_3}{G} = 1$, so $G^{**} = 60$. The equilibrium provides only half as much. Each neighbor stops contributing when her *own* marginal benefit a_i/G equals the marginal cost 1, ignoring the benefit to the others (the Samuelson condition requires the *sum* of marginal benefits to equal the marginal cost). Neighbors 2 and 3 free ride.

> **Grading.** 2 pts: $G^{**} = 60$. 1 pt: comparison. 2 pts: explanation (free riding / Samuelson condition).

(d) Now contributors must satisfy $G = 20$ if they are neighbor 1 or 2, and $G = 10$ if neighbor 3. If neighbor 3 contributed, $G = 10$, but then neighbors 1 and 2 would have $G_{-j} \leq 10 < 20$ and would want to contribute: impossible. So $g_3 = 0$, $G = 20$, and any split of 20 between neighbors 1 and 2 works (each non-contributor faces $G_{-j} = 20 \geq a_j$; each contributor tops up to exactly 20):

$$\{(g_1, 20 - g_1, 0) : \ 0 \leq g_1 \leq 20\} .$$

There is a continuum of equilibria; $G = 20$ in every one of them, but the division of the cost between neighbors 1 and 2 is indeterminate.

Grading. 2 pts: neighbor 3 does not contribute. 3 pts: the full set of equilibria (1 pt if only the two corner equilibria are given). 1 pt: $G = 20$ in all.

Question 5: The Transit Contract (20 points)

(a) Work backward (shares of \$12M; amounts in each period's own dollars).

- Period 3 (city proposes): the union accepts any offer ≥ 0, so the city takes 12 and offers 0.
- Period 2 (union proposes): rejecting gives the city $\delta_C \cdot 12 = 9.6$. The union offers the city 9.6 and keeps 2.4; the city accepts.
- Period 1 (city proposes): rejecting gives the union $\delta_U \cdot 2.4 = 1.2$. The city offers the union 1.2 and keeps 10.8; the union accepts.

SPE: the city proposes $(10.8,\ 1.2)$ in period 1 and it is accepted immediately. (Full strategies: in period 1 the union accepts iff offered at least 1.2; in period 2 the city accepts iff offered at least 9.6; in period 3 the union accepts any offer.)

Grading. 2 pts: period 3. 2 pts: period 2. 2 pts: period 1. 2 pts: acceptance rules/immediate agreement. Using a common discount factor for both parties: at most 4/8.

(b) Stationarity: when the city proposes, the union must be indifferent between accepting $12 - x$ now and rejecting to receive $12 - y$ next period; when the union proposes, the city must be indifferent between y now and x next period:

$$12 - x = \delta_U \, (12 - y), \qquad y = \delta_C \, x.$$

Substituting, $12 - x = \delta_U (12 - \delta_C x)$, so

$$x = \frac{12(1 - \delta_U)}{1 - \delta_C \delta_U} = \frac{12 \cdot 0.5}{0.6} = \ 10 \ , \qquad y = 0.8 \cdot 10 = 8.$$

The period-1 split is \$10M city, \$2M union , accepted immediately.

Grading. 3 pts: two indifference conditions. 3 pts: solving for x and y. 2 pts: period-1 split. Quoting the formula $x = \frac{1-\delta_U}{1-\delta_C\delta_U}$ without derivation: 4/8.

(c) With $\delta_U = 0.75$: $x = \dfrac{12 \cdot 0.25}{1 - 0.6} = 7.5$, so the city gets \$7.5M and the union \$4.5M (up from \$2M). A more patient union loses less by rejecting and waiting, so the city must offer more to secure agreement.

Grading. 2 pts: new split. 2 pts: explanation.